DERIVATIVES AND SHAPE

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Name: Solutions

Critical Numbers

Find the critical numbers of the given function

1. $f(x) = x^3 + 6x^2 - 15x$

$$0 = f'(x) = 3x^2 + 12x - 15$$
$$= 3(x^2 + 4x - 5)$$
$$= 3(x+5)(x-1)$$
$$\Leftrightarrow \boxed{x = 1 \text{ or } x = -5}$$

2. $f(x) = 2x^3 + x^2 + 2x$

$$f'(x) = 6x^2 + 2x + 2$$
$$D = 2^2 - 4(6)(2) = 4 - 48 < 0$$
$$\Rightarrow f'(x) \neq 0, \text{ so no critical points.}$$

3. $f(x) = |3x - 4|$

$$= \begin{cases} 3x - 4 & 4/3 \leq x \\ 4 - 3x & x < 4/3 \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} 3 & 4/3 < x \\ -3 & x < 4/3 \\ \text{undefined} & x = 4/3 \end{cases}$$

$$\boxed{x = 4/3}$$

4. $f(x) = \dfrac{x - 1}{x^2 - \cancel{1}\,x + 1}$

Note:
$$(-1)^2 - 4(1)(1) = 1 - 4 < 0$$
$$\Rightarrow x^2 - x + 1 \neq 0.$$

$$0 = f'(x) = \frac{x^2 - x + 1 - (x-1)(2x-1)}{(x^2 - x + 1)^2}$$
$$= \frac{x^2 - x + 1 - (2x^2 - 3x + 1)}{(x^2 - x + 1)^2}$$
$$= \frac{x^2 - x + 1 - 2x^2 + 3x - 1}{(x^2 - x + 1)^2}$$
$$= \frac{-x^2 + 2x}{(x^2 - x + 1)^2}$$
$$= \frac{-x(x-2)}{(x^2 - x + 1)^2} \Leftrightarrow \boxed{x = 0,\ x = 2}$$

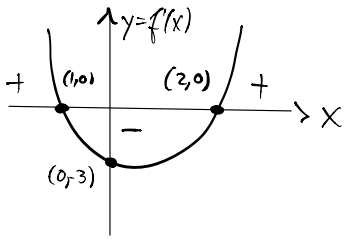
First Derivatives

For each function use the first derivative to find

- the interval(s) where the given function is increasing
- the interval(s) where the given function is decreasing
- the local maximum/minimum values

5. $f(x) = 2x^3 - 9x^2 + 12x - 3$

$f'(x) = 6x^2 - 18x + 12$
$= 6(x^2 - 3x + 2)$
$= 6(x-2)(x-1)$



Increasing: $(-\infty, 1) \cup (2, \infty)$

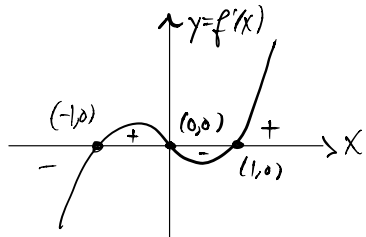
Decreasing: $(1, 2)$

local max when $x = 1$

local min when $x = 2$

6. $f(x) = x^4 - 2x^2 + 3$

$f'(x) = 4x^3 - 4x$
$= 4x(x^2 - 1)$
$= 4x(x+1)(x-1)$



Increasing: $(-1, 0) \cup (1, \infty)$.

Decreasing: $(-\infty, -1) \cup (0, 1)$.

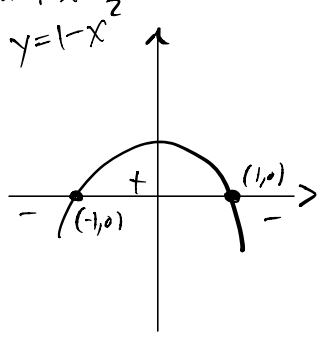
local max when $x = 0$.

local min when $x = -1$ & $x = 1$.

7. $f(x) = \dfrac{x}{x^2+1}$

$$f'(x) = \frac{x^2+1 - x(2x)}{(x^2+1)^2}$$
$$= \frac{x^2+1-2x^2}{(x^2+1)^2}$$
$$= \frac{1-x^2}{(x^2+1)^2} = \frac{(1-x)(1+x)}{(x^2+1)^2}$$

Since $(x^2+1)^2 > 0$, the sign of $f'(x)$ depends only on $1-x^2$:



Increasing: $(-1, 1)$

Decreasing: $(-\infty, -1) \cup (1, \infty)$

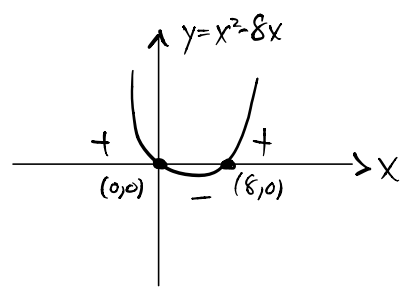
local max when $x = 1$

local min when $x = -1$.

8. $f(x) = \dfrac{x^2}{x-4}$

$$f'(x) = \frac{2x(x-4) - x^2}{(x-4)^2} = \frac{2x^2 - 8x - x^2}{(x-4)^2}$$
$$= \frac{x^2 - 8x}{(x-4)^2} = \frac{x(x-8)}{(x-4)^2}$$

Since $(x-4)^2 \geq 0$, the sign of $f'(x)$ depends only on $x^2 - 8x$



Increasing: $(-\infty, 0) \cup (8, \infty)$

Decreasing: $(0, 8)$

local max when $x = 0$

local min when $x = 8$.

Second Derivatives

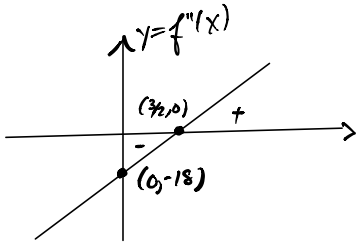
For each function use the second derivative to find

- the interval(s) where the given function is concave up
- the interval(s) where the given function is concave down
- the local maximum/minimum values
- the inflection points

9. $f(x) = 2x^3 - 9x^2 + 12x - 3$

$f'(x) = 6x^2 - 18x + 12$

$f''(x) = 12x - 18 = 6(2x-3)$



Inflection Points: $x = 3/2$

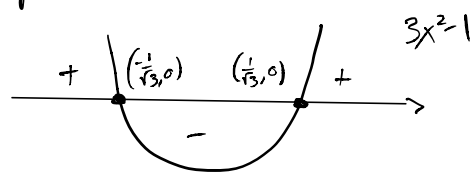
Concave up: $(3/2, \infty)$

Concave Down: $(-\infty, 3/2)$

10. $f(x) = x^4 - 2x^2 + 3$

$f'(x) = 4x^3 - 4x$

$f''(x) = 12x^2 - 4 = 4(3x^2-1) = 0 \iff x^2 = \frac{1}{3} \iff x = \pm\frac{1}{\sqrt{3}}$



Inflection Points: $x = \pm\frac{1}{\sqrt{3}}$

Concave up: $(-\infty, \frac{-1}{\sqrt{3}}) \cup (\frac{1}{\sqrt{3}}, \infty)$

Concave down: $(\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$

11. $f(x) = \dfrac{x}{x^2+1}$

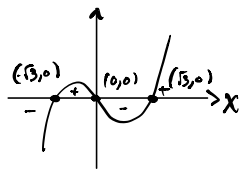
$f'(x) = \dfrac{1-x^2}{(x^2+1)^2}$

$f''(x) = \dfrac{-2x(x^2+1)^2 - (1-x^2)(2)(x^2+1)(2x)}{(x^2+1)^4}$

$= \dfrac{-2x(x^2+1) - 4x(x^2-1)}{(x^2+1)^3} = \dfrac{-2x^3 - 2x + 4x^3 - 4x}{(x^2+1)^3}$

$= \dfrac{2x^3 - 6x}{(x^2+1)^3} = \dfrac{2x(x^2-3)}{(x^2+1)^3} = \dfrac{2x(x-\sqrt{3})(x+\sqrt{3})}{(x^2+1)^3}$

Since $x^2+1>0$ the concavity is determined by $2x^3-6x$



Inflection Points: $x = 0, \pm\sqrt{3}$

Concave up: $(-\sqrt{3}, 0) \cup (\sqrt{3}, \infty)$

Concave down: $(-\infty, -\sqrt{3}) \cup (0, \sqrt{3})$

12. $f(x) = \dfrac{x^2}{x-4}$

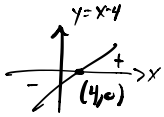
$f'(x) = \dfrac{x^2 - 8x}{(x-4)^2}$

$f''(x) = \dfrac{(2x-8)(x-4)^2 - (x^2-8x)(2)(x-4)}{(x-4)^4}$

$= \dfrac{(2x-8)(x-4) - 2(x^2-8x)}{(x-4)^3}$

$= \dfrac{2x^2 - 16x + 32 - 2x^2 + 16x}{(x-4)^3}$

$= \dfrac{32}{(x-4)^3} \neq 0.$



Inflection Points: None

Concave up: $(4, \infty)$

Concave Down: $(-\infty, 4)$